

Adding, Subtracting, and Multiplying Complex Numbers

Answer Key

Algebra 2

Quick Answers

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|---------------|------------------------------------|
| 1. $7 + 3i$ | 7. 29 |
| 2. $5 - 7i$ | 8. $7 - 24i$ |
| 3. $21 + 12i$ | 9. $1 - i$ |
| 4. $13 + 13i$ | 10. $3 + 2i, 3 - 2i$ |
| 5. -1 | 11. $9 + 7i$ |
| 6. $3 + 3i$ | 12. $x^2 - 10x + 26, 5 + i, 5 - i$ |
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Common wrong answers

7. If they got 21: treated i squared as $+1$, so the last product read 25 minus 4 instead of 25 plus 4

1. Simplify $(3 + 5i) + (4 - 2i)$.

Real parts combine with real parts, imaginary parts with imaginary parts — i behaves exactly like a variable here.

Real: $3 + 4 = 7$. Imaginary: $5i - 2i = 3i$.

$$\boxed{7 + 3i}$$

2. Simplify $(8 - i) - (3 + 6i)$.

Distribute the minus sign over both parts of the second number before combining:

$$8 - i - 3 - 6i$$

Real: $8 - 3 = 5$. Imaginary: $-i - 6i = -7i$.

$$\boxed{5 - 7i}$$

Grading note: $5 + 5i$ means only the real part of the second number was negated.

3. Multiply $3i(4 - 7i)$.

Distribute, then use $i^2 = -1$:

$$3i \cdot 4 - 3i \cdot 7i = 12i - 21i^2 = 12i - 21(-1)$$

The i^2 term becomes a real number, so it moves to the front.

$$\boxed{21 + 12i}$$

4. Multiply $(2 + 3i)(5 - i)$.FOIL first, replace i^2 second:

$$10 - 2i + 15i - 3i^2 = 10 + 13i - 3(-1) = 10 + 13i + 3$$

$$\boxed{13 + 13i}$$

5. Simplify i^{18} .Powers of i repeat every four steps, so divide the exponent by 4 and keep the remainder:

$$18 = 4 \cdot 4 + 2, \text{ remainder } 2.$$

$$\text{So } i^{18} = (i^4)^4 \cdot i^2 = 1^4 \cdot i^2 = i^2.$$

$$\boxed{-1}$$

6. Simplify $(-4 + 7i) + (9 - 7i) - (2 - 3i)$.

Distribute the minus over the third number, then collect:

$$-4 + 7i + 9 - 7i - 2 + 3i$$

$$\text{Real: } -4 + 9 - 2 = 3. \quad \text{Imaginary: } 7i - 7i + 3i = 3i.$$

The imaginary parts of the first two numbers are additive inverses, so they cancel completely; the whole imaginary part of the answer comes from the third number.

$$\boxed{3 + 3i}$$

7. Multiply $(5 + 2i)(5 - 2i)$.

These are conjugates, so the outer and inner products cancel:

$$25 - 10i + 10i - 4i^2 = 25 - 4i^2 = 25 - 4(-1) = 25 + 4$$

The result is real because the $-4i^2$ term turns into $+4$: a complex number times its conjugate is always the real number $a^2 + b^2$.

$$\boxed{29}$$

Grading note: 21 is the diagnostic wrong answer — it means i^2 was treated as $+1$ (or the $-4i^2$ was simply read as -4). Every product problem on this sheet turns on that one substitution.**8. Expand** $(4 - 3i)^2$.

Squaring a binomial is multiplying it by itself, so there are four products, not two:

$$(4 - 3i)(4 - 3i) = 16 - 12i - 12i + 9i^2 = 16 - 24i + 9(-1)$$

$$\boxed{7 - 24i}$$

Grading note: 25 means the student squared each term separately and dropped the middle. $16 + 9$ is not $(4 - 3i)^2$; it is the square of the modulus.**9. Simplify** $i^{43} + i^{20}$.

Take each exponent mod 4.

$$43 = 4 \cdot 10 + 3, \text{ so } i^{43} = i^3 = -i.$$

$$20 = 4 \cdot 5 + 0, \text{ so } i^{20} = i^0 = 1.$$

Adding: $-i + 1$, which in standard form is written real part first.

$$\boxed{1 - i}$$

10. Solve $x^2 - 6x + 13 = 0$ over the complex numbers.

The discriminant is negative, which is exactly when the solutions are a conjugate pair:

$$\Delta = (-6)^2 - 4(1)(13) = 36 - 52 = -16$$

$$x = \frac{6 \pm \sqrt{-16}}{2} = \frac{6 \pm 4i}{2}$$

Divide both parts of the numerator by 2.

$$\boxed{x = 3 + 2i \text{ or } x = 3 - 2i}$$

Check by multiplying: $(3 + 2i)^2 - 6(3 + 2i) + 13 = 5 + 12i - 18 - 12i + 13 = 0 \checkmark$.

11. Multiply $(2 + i)(3 - 2i)(1 + i)$.

Do it in two stages, returning to standard form after each stage — carrying an unsimplified i^2 into the next product is where this problem goes wrong.

Stage 1: $(2 + i)(3 - 2i) = 6 - 4i + 3i - 2i^2 = 6 - i + 2 = 8 - i$.

Stage 2: $(8 - i)(1 + i) = 8 + 8i - i - i^2 = 8 + 7i + 1$.

$$\boxed{9 + 7i}$$

12. Synthesis: expand $(x - (5 + i))(x - (5 - i))$ and state its zeros.

Group the real part with x and treat the pair as a difference of squares:

$$((x - 5) - i)((x - 5) + i) = (x - 5)^2 - i^2 = (x - 5)^2 + 1$$

Expand the square: $x^2 - 10x + 25 + 1$.

$$\boxed{x^2 - 10x + 26}$$

The factors were built from x minus each root, so the zeros are exactly the two numbers that were subtracted.

$$\boxed{x = 5 + i \text{ or } x = 5 - i}$$

The coefficients came out real because the two roots are conjugates: their sum $(5 + i) + (5 - i) = 10$ and their product $(5 + i)(5 - i) = 26$ are both real, and those are precisely the coefficients the expansion produces.